

User Manual: Investment Metrics Explained

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This manual covers every metric, formula, and setting available in the CRE Deal Analyzer. It includes both the original residential/commercial metrics and the new CRE-grade features added in Version 2: Lease Type modeling, Tenant Credit tiers, Broker Fee impact, and Rent Bump schedules.

1. Core Investment Metrics

Net Present Value (NPV)

NPV answers: *How much is this stream of future profits worth today?*

Rule	Meaning
NPV > 0	Investment is profitable at your required rate
NPV = 0	Return exactly equals your discount rate (= IRR)
NPV < 0	Investment destroys value at that discount rate

Formula: $NPV = \sum CF_t / (1 + r)^t - \text{Initial Investment}$

Internal Rate of Return (IRR)

IRR is the annualized return rate where NPV equals zero. We publish two versions:

IRR Type	What It Measures
IRR (Operational)	Return from annual cash flows only — rent minus expenses minus mortgage. Excludes a
IRR (Total incl. Sale)	Adds terminal sale proceeds (or modeled appreciation at exit) to the final year cash flow.

Tip: Use Operational IRR to compare rental efficiency across deals. Use Total IRR to evaluate full lifecycle return including your exit.

Cash-on-Cash Return (CoC)

Measures annual pre-tax cash flow relative to initial cash invested.

Formula: $CoC = (\text{Annual Pre-Tax Cash Flow} / \text{Initial Cash Invested}) \times 100$

CoC Grade	Range	Interpretation
A	≥ 15%	Exceptional — rare in most markets

B	12–15%	Strong performer
C	9–12%	Solid, market-rate return
D	6–9%	Marginal — review assumptions
F	< 6%	Negative or very thin — reconsider

Cap Rate

Capitalization rate measures a property's income yield independent of financing.

Formula: $\text{Cap Rate} = (\text{NOI} / \text{Purchase Price}) \times 100$

NOI (Net Operating Income) = Effective Rent – Operating Expenses. Cap rate does not include mortgage payments — it is a property-level metric, not an equity metric. Tenant credit tier adjustments are applied on top of the base cap rate (see Section 4).

Equity Multiple

Formula: $\text{Equity Multiple} = \text{Total Cash Inflows} / \text{Total Cash Invested}$

Measures how many times your original equity has grown over the hold period. Example: invest \$100K, receive \$220K total (rent + sale) = 2.2x equity multiple.

DSCR — Debt Service Coverage Ratio

Formula: $\text{DSCR} = \text{NOI} / \text{Annual Debt Service (mortgage payments)}$

DSCR	Signal
$\geq 1.25\times$	Lender-approved — standard minimum for most commercial loans
$1.0\text{--}1.25\times$	Borderline — some lenders accept, higher rates likely
$< 1.0\times$	Below breakeven — NOI does not cover debt service

ROI (Return on Investment)

Formula: $\text{ROI} = (\text{Total Return} / \text{Initial Investment}) \times 100$

Total return includes: appreciation value, mortgage principal paydown, and net cash flow. ROI is cumulative and grows each year. Year 1 ROI often equals CoC when appreciation and principal paydown are minimal.

2. Cash Flow & Multi-Year Projections

How Cash Flow Is Calculated

Cash Flow = Annual Effective Rent – Operating Expenses – Mortgage Payments

Effective Rent = Scheduled Rent × (1 – Economic Vacancy Rate). This is the rent actually collected after vacancy and credit loss.

Economic Vacancy vs. Vacancy Rate

The app uses **Economic Vacancy** — the CRE-standard term. It captures both physical vacancy (empty units) AND credit loss (non-paying tenants). This is more conservative and accurate than a simple vacancy rate, particularly for multifamily and net lease assets.

Tip: If unsure, set Economic Vacancy to 5% for stabilized assets, 8–10% for value-add.

The Multi-Year Cash Flow Projection Chart

The chart plots three lines over your hold period:

Line	Color	What It Shows
Multi-Year Cash Flow (\$)	Blue	Annual net cash flow after all expenses and debt service
Pro Forma Rent (\$/yr)	Orange	Projected gross annual rent growing at your rent growth rate (plus any bump schedule)
ROI (%)	Green	Cumulative return on equity including appreciation, compounding each year

Left Y-axis shows dollar values (cash flow and rent). Right Y-axis shows ROI percentage. This dual-axis design is your competitive advantage — no other tool in this price range shows all three trajectories simultaneously.

Hold Period (Years)

The number of years you plan to own the property before sale. Affects IRR Total, Equity Multiple, ROI, and the terminal sale value used in all exit-based calculations.

3. Lease Type Modeling [NEW in v2]

The lease structure determines who pays operating expenses — the landlord or the tenant. This fundamentally changes NOI, cap rate, and cash flow calculations.

Lease Type	How Expenses Are Handled	Typical Asset Classes
Gross	Landlord pays ALL operating expenses (taxes, insurance, maintenance, utilities, etc.)	Single-family home, office space
Modified Gross	Expenses are split — landlord pays base expenses, tenant pays overhead or specific items.	Multi-unit rental, office space
NNN (Triple Net)	Tenant pays ALL operating expenses: property taxes, insurance, and maintenance.	Single-unit rental, Starbucks, Walgreens, Dollar General

How NNN affects your numbers: When you select NNN, the OpEx field is overridden to \$0. This increases NOI and therefore cap rate, cash flow, and DSCR — all improving. This accurately reflects the economics of a true NNN lease where the landlord's only variable is rent collection.

Tip: For NNN deals, verify the lease terms. Some NNN leases exclude roof and structure from tenant responsibility — in that case, retain a small OpEx reserve (e.g. \$100–200/mo).

4. Tenant Credit Rating [NEW in v2]

In net lease investing, the tenant's creditworthiness is one of the most important valuation drivers. Investment-grade tenants command a premium (lower cap rate = higher price). Non-rated or franchisee tenants trade at a discount (higher cap rate = lower price).

Credit Tier	Definition	Cap Rate Adjustment	Example Tenants
Investment Grade	S&P rated BBB- or higher (or Moody's Baa3 or higher) (or higher)	-0.75% (tighter)	Walgreens, CVS, Chase, Dollar
Non-Rated	No public credit rating. Private companies or smaller operators (baseline)	0.00%	Regional retailers, independent operators
Franchisee	Franchise operator running a branded location	+0.50% (wider yield)	McDonald's franchisee, LaSalle, Burger King, Papa

How the adjustment works: The cap rate displayed adds or subtracts the tier adjustment from your property's base cap rate (NOI / Price). This reflects how the market would price the asset given tenant quality. It is a display and analysis tool — the underlying cash flows do not change based on credit tier.

Tip: Cap rate adjustments of 50–150 bps for credit tier are consistent with published Marcus & Millichap and CBRE Net Lease cap rate surveys. Exact spreads vary by market, asset type, and interest rate environment.

5. Broker Fee Modeling [NEW in v2]

The broker fee input allows you to model the cost of sale — the commission paid to the selling and buying brokers at exit — and see its precise impact on your IRR and equity multiple.

How It Works

$$\text{Broker Fee (\$)} = \text{Purchase Price} \times \text{Broker Fee \%}$$

This dollar amount is subtracted from the terminal sale value in the final year of your hold period. It reduces the net proceeds you receive at exit, which flows directly into the IRR (Total incl. Sale) and Equity Multiple calculations.

Typical Broker Fee Ranges	Asset Type
1.5–2.5%	Large commercial assets (\$5M+)
2.0–3.0%	Mid-market net lease (\$1M–\$5M)
3.0–4.0%	Small commercial / residential investor deals

Tip: Standard net lease retail (Andy Sabherwal's market at Matthews) typically runs 2–3% total commission. Set to 2.5% as a reasonable default for most CRE deals.

Why This Matters

On a 10-year hold, a 2.5% broker fee on a \$1M property reduces your exit proceeds by \$25,000. This can compress IRR by 0.3–0.8% depending on your hold period. Including it gives you a more honest picture of actual investor returns.

6. Rent Bump & Escalation Schedule [NEW in v2]

Rent bump schedules are contractual rent increases built into a lease — common in net lease retail where rent growth is fixed and predictable rather than market-driven.

Rent Bump % and Bump Every N Years

Setting	What It Does
Rent Bump (%)	The percentage increase applied to base rent at each bump interval. Example: 10% bump
Bump Every N Years	How often the bump occurs. Example: every 5 years means year 5, year 10, year 15 rec

How It Interacts With Rent Growth Rate

In years when a scheduled bump occurs, the bump percentage replaces the annual rent growth rate for that year. In all other years, the annual rent growth rate applies normally. This accurately models a typical NNN lease where rent is contractually fixed between bumps.

Year	Rent Growth Applied	Example (Base \$60K/yr, 10% bump every 5 yr, 3% annual growth)
Year 1	Base rent	\$60,000
Year 2	3% growth	\$61,800
Year 3	3% growth	\$63,654
Year 4	3% growth	\$65,564
Year 5	10% bump	\$66,000 (10% of original \$60K base)
Year 6	3% growth	\$67,980
Year 10	10% bump	\$72,600 (second bump cycle)

Tip: Dollar General, Family Dollar, and most single-tenant net lease deals use 10% bumps every 5 years. Starbucks and some QSR tenants use 10% every 5 or fixed 1.5–2% annual escalations. Match this setting to the actual lease terms.

7. Operating Expenses (OpEx)

Monthly operating expenses include all recurring costs the **landlord** is responsible for:

Expense Category	Notes
Property Taxes	Largest component in most markets. Verify with county assessor.
Property Insurance	Hazard + liability. Typically \$100–400/mo for residential, higher for commercial.
Maintenance & Repairs	Budget 0.5–1% of property value annually for residential. NNN leases shift this to tenant.
HOA Fees	Where applicable. Fixed monthly charge.
Property Management	If third-party managed: typically 8–12% of gross rent.
Miscellaneous	Landscaping, pest control, professional fees, etc.

Tip: If unsure of total OpEx, estimate 25–30% of gross monthly rent for residential. For NNN leases, set OpEx to \$0 or a small structural reserve (\$100–200/mo).

8. IRR (Operational) vs IRR (Total incl. Sale)

IRR Type	Cash Flows Used	Best For
IRR (Operational)	[-Down Payment, CF1, CF2, ..., CFn] — no sale proceeds	Comparing rental efficiency. Lender underwriting.
IRR (Total incl. Sale)	[-Down Payment, CF1, CF2, ..., CFn + Sale Value] — includes sale	Blended rate of return. Investor decision-making.

Illustrative Example:

Input	Value
Down Payment	\$40,000
Annual Net Cash Flow	\$3,600
Exit Proceeds (Year 5)	\$60,000
IRR (Operational)	~8.7%
IRR (Total incl. Sale)	~17.7%

9. Access Control: Password & PIN Gating

The platform supports two-layer access control to protect the deal analysis environment:

Layer 1 — App Password

A single shared password grants entry to the app. This is suitable for broad distribution to agents, investors, and prospects. The password is case-sensitive and should be shared only with intended users.

Layer 2 — Per-User PIN

Individual PINs allow granular access control per user. Each user receives a unique PIN which can be revoked independently without changing the main app password. PINs are stored securely in environment variables using double-underscore notation (e.g. `pins__username=1234` on Render).

Access Layer	Scope	Where Configured
App Password	All users — broad access gate	.streamlit/secrets.toml or Render env var APP_PASSWORD
User PIN	Per-user — individual identity	<code>pins__username=NNNN</code> in secrets.toml or Render dashboard

Tip: PINs on Render use double-underscore notation to represent nested config: `pins__masoud=7741` maps to `pins.masoud = 7741` in the app.

B2B Embed Access

When the widget is embedded via `iframe` in a partner platform (OneSource, Brevitas, bidCRM), end users access the analysis tool directly — no login required on their side. Access control is managed at the partner integration level, not the widget level. API-based integrations authenticate via the X-API-Key header.

10. API Integration for CRM Platforms

The CRE Deal Analyzer exposes a REST API endpoint for integration into third-party platforms. This allows broker CRMs, marketplaces, and workflow tools to trigger deal analysis programmatically and receive structured JSON results.

Endpoint

POST /analyze-deal

Header: X-API-Key: your_api_key

Required Input Fields

Field	Type	Example	Notes
<code>purchase_price</code>	float	800000	Acquisition price in dollars
<code>monthly_rent</code>	float	6500	In-place rent per month
<code>down_payment_pct</code>	float	25	Equity contribution %
<code>mortgage_rate</code>	float	6.75	Annual interest rate %

mortgage_term	int	30	Amortization in years
monthly_expenses	float	0	Set to 0 for NNN leases
vacancy_rate	float	5	Economic vacancy %
appreciation_rate	float	3	Annual appreciation %
rent_growth_rate	float	3	Annual rent growth %
time_horizon	int	10	Hold period in years

Integration Models

Model	How It Works	Best For
Shareable URL	Partner passes inputs via URL params. App generates report on the spot. Analysis page in browser. No API key required.	One-time, ad-hoc analysis. No integration needed.
iframe Embed	Single line of HTML embeds the full interactive widget in partner's page. No API key required.	Embedding the widget on a partner's website for on-demand use.
API + Partner UI	Partner calls /analyze-deal API, receives JSON, renders results in their own UI. No API key required.	Large platforms with dedicated developer resources.